

REVIEW



# Cost-effectiveness analyses of 20-valent pneumococcal conjugate vaccine in children and adults: A narrative review

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## ABSTRACT

Many pneumococcal vaccines have been developed. Economic evaluation provides a framework to inform decision-making. Cost-effectiveness analysis (CEA) studies on 20-valent pneumococcal conjugate vaccine (PCV20) have been published. We introduce the concept of health economics and the basic framework for CEAs. We present a narrative literature review of recent CEA studies involving the PCV20. We also provide guidance for interpreting vaccine-related CEAs. We included 26 CEAs (17 in adults, nine in children) from 15 countries in this review. Eight of the pediatric studies and 11 of the adult studies concluded that PCV20 strategy would be cost-saving compared with existing vaccine(s), in at least one comparison. CEA is a modeling method based on existing data but includes a degree of uncertainty. It is therefore important to interpret the results of a CEA study with an understanding of the features and limitations of the analysis.

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## Introduction

*Streptococcus pneumoniae*, a cause of pneumonia, may lead to invasive pneumococcal disease and death, with a heavy global mortality burden.<sup>1</sup> Pneumococcal infections may cause otitis, sinusitis, pneumonia and meningitis in severe cases.<sup>1</sup> Many new pneumococcal vaccines have been developed in recent decades.<sup>1,2</sup> There are currently two types of vaccine available: pneumococcal polysaccharide vaccine (PPSV) and pneumococcal conjugate vaccine (PCV). Since the release of the 23-valent pneumococcal polysaccharide vaccine (PPSV23), the 7-valent PCV (PCV7), 10-valent PCV (PCV10), 13-valent PCV (PCV13), 15-valent PCV (PCV15), 20-valent PCV (PCV20), and 21-valent PCV (PCV21) have been developed and released in some countries.<sup>3,4</sup>

In the United States (US), PCV 20 and PCV15 were licensed in 2021 for adults aged  $\geq 18$  years by the Food and Drug Administration (FDA).<sup>5</sup> In 2024, the FDA approved PCV21 for adults aged  $\geq 18$  years, and the US Advisory Committee on Immunization Practices recommended PCV21 as an option for adults aged  $\geq 19$  years who currently have a recommendation to receive a dose of PCV.<sup>6</sup> In Europe, PCV20 was authorized by the European Medicines Agency in 2022.<sup>7</sup>

In 2023, PCV20 was approved for children aged 6 weeks through 17 years by the US FDA, and the Advisory Committee on Immunization Practices recommended PCV20 for children as an option to PCV15.<sup>8</sup> In Canada, PCV20 was authorized for children 6 weeks to 17 years old in 2023; in 2024, the Public Health Agency of Canada released the National Advisory Committee on Immunization's public health recommendations on the use of either PCV20 or PCV15 in children.<sup>9</sup>

With continuous vaccine development, governments and policymakers have to decide whether individuals should be encouraged

to take up a new vaccine as part of a national immunization program (NIP) or whether a new immunization program for specific target populations should be launched instead.<sup>3,10</sup> As such, certain concepts used in economics may offer useful information for stakeholders in the planning and provision of new interventions.<sup>10,11</sup> *Economic evaluation* refers to the evaluation of alternative policies or interventions; in the context of vaccines, this refers to investigating the "value for money" of a new vaccination program or expansion of a vaccination service in terms of both health outcomes and health care resources.<sup>10–12</sup> As economic evaluations of PCV20, several cost-effectiveness analysis (CEA) studies have been published; however, few CEAs of the newer PCV21 have been published at the time of reviewing the literature.<sup>13</sup>

In this article, we first introduce the basic conceptual framework for CEAs. Next, we present the findings of a narrative literature review (not a systematic review) of PCV20 against pneumococcal disease. We also describe how each CEA was conducted, and we provide guidance for readers who are unfamiliar with these studies in how to interpret CEAs related to vaccines, based on the analytical framework in the reviewed articles.

## Conceptual framework for conducting cost-effectiveness research

The main economic evaluation methods comprise cost analysis, cost-benefit analysis (CBA), CEA, and cost-utility analysis (CUA) (Table 1). A cost analysis only considers cost comparisons; in the other analyses, both cost and consequences are compared.<sup>10,12</sup> Here, we provide guidance on CEA, which is often used in health-care settings.

**Table 1.** Types of economic evaluation.

|                                   | Measurement of costs | Measurement of consequences             |
|-----------------------------------|----------------------|-----------------------------------------|
| Cost analysis                     | Monetary units       | Not applicable                          |
| Cost-benefit analysis (CBA)       | Monetary units       | Monetary units                          |
| Cost-effectiveness analysis (CEA) | Monetary units       | Natural units (e.g., life-years gained) |
| Cost-utility analysis (CUA)       | Monetary units       | QALYs*                                  |

\*Quality-adjusted life years.

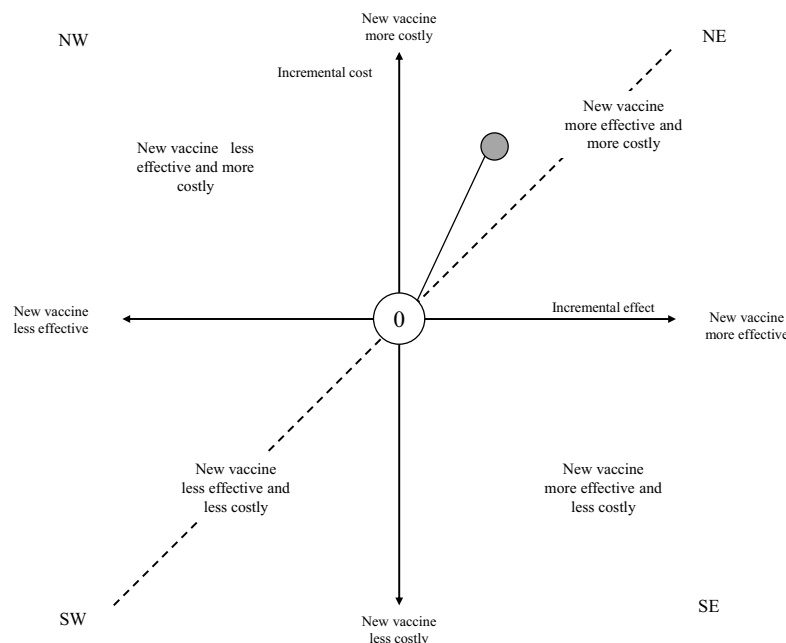
In CEAs, health outcomes, e.g., life-years gained, can be set according to the intended purpose. The effects of a health intervention may be measured in life-years and quality of life. CUA is a variant of CEA in which the quality-adjusted life-year (QALY) is used as a measure of an intervention's effectiveness, to capture its multifaceted effects on health states. The CUA using QALYs is recommended in many guidelines, and it is widely used for economic evaluation in many studies in the field of health research.<sup>10,14</sup>

The comparator set in the CEA depends on the purpose of the economic evaluation and it impacts on the results of the analysis and the interpretation of the result. Potential options for the CEA comparator can vary according to the purpose and background in each setting. The comparator scenarios for evaluation of a new vaccine might include the recommended vaccine schedule, the available vaccines, or no vaccination.<sup>14</sup> The outcome of the CEA is expressed as the incremental cost-effectiveness ratio (ICER), e.g., the unit cost to gain 1 QALY. The ICER is calculated using the following formula:

$$\begin{aligned}
 &\text{Incremental Cost effectiveness Ratio (ICER)} \\
 &= \frac{\text{Cost}_{\text{new vaccine program}} - \text{Cost}_{\text{existing vaccine program}}}{\text{Effect}_{\text{new vaccine program}} - \text{Effect}_{\text{existing vaccine program}}} \\
 &= \frac{\text{Incremental Cost}}{\text{Incremental Effect}}
 \end{aligned}$$

For this formula, we assume that a new vaccine becomes available. After calculating the respective costs and effects for both the new vaccine and the existing vaccine, the ICER is calculated as the difference in costs divided by the difference in effects. Note that the cost here includes not only vaccine administration but also treatment of the disease and the related costs. Incremental cost and incremental effectiveness can be plotted as a cost-effectiveness plane, as shown in Figure 1.

The central point in Figure 1 (zero) indicates the existing vaccine option. The incremental cost of the new vaccine relative to the existing vaccine is plotted on the y-axis, and the incremental effectiveness of the new vaccine relative to the comparator is plotted on the x-axis. The cost is measured in the country's currency, and QALY is widely used as the effectiveness measure.<sup>10,12</sup> The outcome of a CEA is drawn at one point (gray circle) in the cost-effectiveness plane, and the slope of the straight line between the gray circle and the central point of the figure indicates the ICER.<sup>10–12</sup> In Figure 1, the dashed line represents the cost-effectiveness threshold. The interpretation is given as follows<sup>10–12,14,15</sup>: if the new vaccine (gray circle) is indicated in the southeast (SE) quadrant of the graph, then it is likely to be more effective and less costly compared with the existing vaccine, and is described as “dominant.” In this case, the choice between vaccines is clear. If it is plotted in the northwest (NW) quadrant, then the new vaccine is likely to be less effective and more costly compared with the existing vaccine, and the choice between the



**Figure 1.** The cost-effectiveness plane and cost-effectiveness threshold. The slope of the straight line: ICER, Incremental cost-effectiveness ratio. Dashed line: cost-effectiveness threshold. Gray circle: intervention of interest: new vaccine.

vaccines is clear, too. The northeast (NE) and southwest (SW) quadrants are “trade-off” cases; the SW offers cost savings at the expense of some health benefits, and the NE depicts a common situation in which the new vaccine is more effective, but also more costly. In the latter case, the choice as to whether the new vaccine should be included in a NIP is more difficult: the slope of the straight line from the gray circle to the center depends on both the effectiveness measure and the cost measure (Figure 1).

### Threshold

To provide guidance on whether the new vaccine is cost-effective, the cost-effectiveness threshold can be used.<sup>11</sup> The cost-effectiveness threshold is sometimes described as the “willingness-to-pay threshold”<sup>14,15</sup> and is the maximum amount of money that society is willing to pay for gaining 1 QALY. Using a cost-effectiveness threshold per health benefit gained, such as cost per QALY, we can consider whether the intervention – a new vaccine – is cost-effective.<sup>11</sup> Researchers should use local thresholds recommended in country-specific guidelines.<sup>16</sup> For instance, in the United Kingdom (UK), the threshold is explicit.<sup>11,17</sup> However, in the US, there is no exact threshold that is commonly recognized.<sup>11</sup> Moreover, many countries do not have local thresholds. In this case, a threshold that is not based on gross domestic product per capita is recommended.<sup>11,14</sup> The transparency and account of the threshold applied are crucial.<sup>11,16</sup>

### Uncertainty

As a feature of CEA, parameters are estimated and modeled using the best existing data available from different sources.<sup>10</sup> Also, some parameters are assumed because existing or reliable data do not exist. Therefore, the uncertainty in each parameter as well as in point estimates should be evaluated.<sup>11</sup> The calculation of uncertainty is known as sensitivity analysis. The range or level for a sensitivity analysis should be defined and the results reported in the study.<sup>10</sup> To specify the uncertainties, a deterministic sensitivity

analysis (DSA) and probabilistic sensitivity analysis (PSA) are usually undertaken.<sup>11</sup> In a DSA, one or more parameters are manually changed and how the results are affected by the change are then investigated. DSAs can be either one-way or multi-way: a one-way sensitivity analysis measures the impact on the results of varying a single parameter while the remainder are kept constant. A multi-way sensitivity analysis measures the impact on the results of changing the value of several parameters at the same time. In a PSA, the joint uncertainty in all parameters varies within a certain range and is input randomly and simultaneously by specifying the distributions and iterations of the model.<sup>12,18</sup> The results are then depicted as a plot of repeated numbers.<sup>14</sup> Figure 2 presents an example of PSA output. Scenario analysis is measures the impact on the results of varying the scenario itself in the base case.<sup>10,14</sup>

Detailed guidance and methodology for how to carry out a CEA are available in various publications<sup>10–12,15,18,19</sup> and the World Health Organization (WHO) also provides standardized guidance for conducting economic evaluations of vaccine programs<sup>14</sup>; standardized country-specific guidelines for conducting economic evaluations are also available. Furthermore, the Consolidated Health Economic Evaluation Reporting Standard was published in 2013,<sup>20</sup> and later revised as Consolidated Health Economic Evaluation Reporting Standard 2022.<sup>21</sup>

## Review of the cost-effectiveness of pneumococcal vaccines

### Search and keywords

In October 2024, we conducted a literature search of papers published in journals using PubMed, which is widely accessible. The search keywords were: “PCV20/20-valent pneumococcal conjugate vaccine” and “cost-effectiveness analysis/economic evaluation/cost-utility analysis.” We excluded CBA and cost minimization analysis study designs, as well as studies of specific/limited populations and catch-up programs. We included 26 CEA studies (17 among adults, nine among children) from 15 countries in this review (Table 2): Argentina

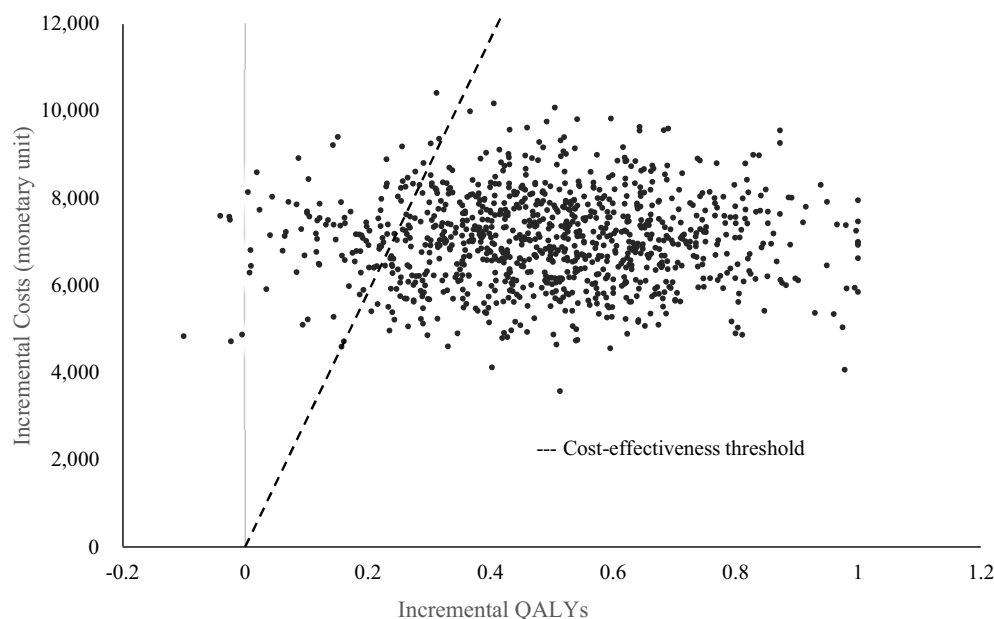


Figure 2. Cost-effectiveness scatter plot.

**Table 2.** The results of CEA studies of PCV20 from 15 countries for review.

| Country     | Children                                |                                                                                                                                                                                                                                                                        |                                             |          | Adults                                        |                                                                                                                                                   |                                                                                                          |          |
|-------------|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|----------|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------|
|             | Authors                                 | Comparisons                                                                                                                                                                                                                                                            | Results                                     | Ref      | Authors                                       | Comparisons                                                                                                                                       | Results                                                                                                  | Ref      |
| Argentina   | Rey-Ares et al. (2024)                  | PCV20 (2 + 1) vs. PCV13 (2 + 1) vs. PCV15 (2 + 1)                                                                                                                                                                                                                      | Cost-saving                                 | 22       | Marbaix et al. (2023)                         | PCV20 vs. 1) no vaccination 2) PCV15+PPSV23                                                                                                       | 1) Cost-effective<br>2) Cost-saving                                                                      | 23       |
| Belgium     | –                                       |                                                                                                                                                                                                                                                                        |                                             |          | –                                             |                                                                                                                                                   |                                                                                                          |          |
| Canada      | Lytle et al. (2023)                     | PCV20 (2 + 1) vs. PCV13 (2 + 1) vs. PCV15 (2 + 1)                                                                                                                                                                                                                      | Cost-saving                                 | 24       | Olsen et al. (2022)                           | PCV20 vs. PPSV23                                                                                                                                  | Cost-saving                                                                                              | 25       |
| Denmark     | –                                       |                                                                                                                                                                                                                                                                        |                                             |          | Kühne et al. (2023)                           | PCV20 vs. current strategy <sup>1</sup>                                                                                                           | Cost-saving                                                                                              | 28       |
| Germany     | Huang et al. (2024)<br>Ta et al. (2024) | PCV20 (3 + 1) vs. PCV15 (2 + 1)<br>PCV20 (3 + 1) vs. PCV13 (2 + 1) vs. PCV15 (2 + 1)                                                                                                                                                                                   | Not cost-effective<br>Cost-saving           | 26<br>27 | –                                             |                                                                                                                                                   |                                                                                                          |          |
| Greece      | –                                       |                                                                                                                                                                                                                                                                        |                                             |          | Gourzoulidis et al. (2023)                    | PCV20 vs. PCV15 vs. PCV15+PPSV23                                                                                                                  | Cost-saving                                                                                              | 29       |
| Italy       | –                                       |                                                                                                                                                                                                                                                                        |                                             |          | Restivo et al. (2023)                         | PCV20 vs. PCV15+PPSV23                                                                                                                            | Cost-saving                                                                                              | 30       |
| Japan       | Shinjo et al. (2024)                    | PCV20 (3 + 1) vs. PCV13 (3 + 1) vs. PCV15 (3 + 1)                                                                                                                                                                                                                      | Cost-saving                                 | 31       | Nakamura et al. (2024)                        | PCV20 vs. PPSV23 vs. PCV15                                                                                                                        | Cost-saving                                                                                              | 32       |
| Netherlands | –                                       |                                                                                                                                                                                                                                                                        |                                             |          | Hoshi et al. (2022)<br>de Boer et al. (2024)  | PCV20 vs. PPSV23<br>PCV20 vs. PCV15, PPSV23(x3), PCV15 +PPSV23(x3)                                                                                | Cost-saving<br>PCV20, cost-saving or cost-effective (depends on comparisons and vaccination in children) | 33<br>34 |
| Norway      | –                                       |                                                                                                                                                                                                                                                                        |                                             |          | Malene et al. (2023)                          | PCV20 vs. no vaccination<br>PCV20+PPSV23(x3) vs. PCV20                                                                                            | Cost-saving<br>PCV21, cost-saving                                                                        | 35       |
| South Korea | Kang et al. (2024)                      | PCV20 (3 + 1) vs. PCV13 (3 + 1)                                                                                                                                                                                                                                        | Cost-saving                                 | 36       | Kang et al. (2024)                            | PCV20 vs. PPSV23                                                                                                                                  | Cost-effective                                                                                           | 37       |
| Spain       | –                                       |                                                                                                                                                                                                                                                                        |                                             |          | Cantarero et al. (2023)                       | PCV20 vs. PCV15+PPSV23                                                                                                                            | Cost-saving                                                                                              | 38       |
| Sweden      | Fridh et al. (2024)                     | PCV20 (2 + 1) vs. PCV15 (2 + 1)                                                                                                                                                                                                                                        | Cost-saving                                 | 39       | –                                             |                                                                                                                                                   |                                                                                                          |          |
| UK          | Wilson et al. (2023)                    | 1) PCV20 (1 + 1) vs. PCV13 (1 + 1)<br>2) PCV20 (1 + 1) vs. PCV15 (1 + 1)<br>3) PCV20 (1 + 1) vs. PCV15 (2 + 1)<br>4) PCV20 (2 + 1) vs. PCV20 (1 + 1)<br>5) PCV20 (2 + 1) vs. PCV13 (1 + 1)<br>6) PCV20 (2 + 1) vs. PCV15 (1 + 1)<br>7) PCV20 (2 + 1) vs. PCV15 (2 + 1) | 1–3), 5–7) Cost-saving<br>4) Cost-effective | 40       | Danelian et al. (2024)                        | PCV20 vs. no vaccination                                                                                                                          | Cost-effective                                                                                           | 41       |
| US          | Rozenbaum et al. (2024)                 | PCV20 (3 + 1) vs. PCV13 (3 + 1) vs. PCV15 (3 + 1)                                                                                                                                                                                                                      | Cost-saving                                 | 43       | Mendes et al. (2022)<br>Wateska et al. (2024) | PCV20 vs. PPSV23<br>PCV20 vs. no vaccination                                                                                                      | Cost-saving<br>Not cost-effective                                                                        | 42<br>44 |
|             |                                         |                                                                                                                                                                                                                                                                        |                                             |          | Altawalbeh et al. (2024)                      | PCV20 <sup>2</sup> vs. PCV15+PPSV23 <sup>1</sup><br>vs. current PCV20 <sup>3</sup><br>vs. current PCV15+PPSV23 <sup>3</sup><br>vs. no vaccination | Potentially economically viable                                                                          | 45       |
|             |                                         |                                                                                                                                                                                                                                                                        |                                             |          | Wateska et al. (2023)<br>Smith et al. (2021)  | PCV21 vs. PCV20<br>PCV20 vs. current recommendation <sup>4</sup><br>PCV20+PPSV23 vs. PCV20                                                        | PCV21, economically favorable<br>Not cost-effective                                                      | 13<br>46 |

<sup>1</sup>+ means 'followed by'.<sup>2</sup>Age/risk-dependent use of PPSV23 or PCV13+PPSV23 with PPSV23 revaccination.<sup>3</sup>At ages 50 and 65 yrs.<sup>4</sup>For all ≥65 yrs and for younger adults with a high risk of pneumococcal disease.<sup>5</sup>PPSV23+optional PCV13 after clinical shared decision making.

(pediatric 1<sup>22</sup>, adult 0), Belgium (0, 1<sup>23</sup>), Canada (1<sup>24</sup>, 0), Denmark (0, 1<sup>25</sup>) Germany (2<sup>26,27</sup>, 1<sup>28</sup>) Greece (0, 1<sup>29</sup>), Italy (0, 1<sup>30</sup>) Japan (1<sup>31</sup>, 2<sup>32,33</sup>) Netherlands (0, 1<sup>34</sup>) Norway (0, 1<sup>35</sup>) South Korea (1, 3<sup>36</sup>, 1<sup>37</sup>) Spain (0, 1<sup>38</sup>) Sweden (1, 3<sup>39</sup>, 0), the UK (1<sup>40</sup>, 2<sup>41,42</sup>) and the US (1<sup>43</sup>, 4, 13,44–46) Except for Argentina, these countries are classified as high-income countries, according to the World Bank's 2024 classification.<sup>47</sup>

### **Current vaccine schedules for pneumococcal disease in countries conducting CEA studies**

To optimize vaccination coverage, governments plan NIPs to sustain routine immunization, and every vaccine schedule is then aligned to target population, age, and health status.<sup>11</sup> We therefore searched the current pneumococcal vaccination schedules for the 15 countries featured in the literature search results to capture details of the current landscape<sup>48–58</sup> (Table 3). Note that the schedules for routine vaccination programs varied between countries. For pediatric vaccination, PCV10 (one country), PCV13 (seven countries), PCV15 (six countries), and PCV20 (three countries) were included in current schedules (with the option to choose in some countries); however, vaccines of unknown valency in schedules were not counted. In some countries, individuals could choose the vaccine valency they wish to receive. Furthermore, the number of doses differed between countries; for example, 1 + 1, 2 + 1, 3 + 1 (where “+” represents a booster vaccine), even if the same vaccine was scheduled. For adults, PCV13 (two), PCV15 (one), PCV20 (six), PCV21 (one), and PPSV23/PPV23 (eight) were scheduled (including with the option to choose or booster vaccine). The target age stipulated in the schedule also varied in these countries (e.g., ≥65 years, or ≥60 years, or ≥75 years).

### **Results of CEA studies of cost-effectiveness for pneumococcal vaccines in children**

Of the nine pediatric studies, eight concluded that the PCV20 strategy would cost less and gain more (i.e., would be cost-saving) compared with either PCV15 or PCV13, although the vaccination frequency (number of doses) differed among the studies. Only one study concluded that PCV20 (3 + 1) was not cost-effective compared with PCV15 (described as V114 in the study) (2 + 1)<sup>26</sup> (Table 2).

### **Results of CEA studies of cost-effectiveness for pneumococcal vaccines in adults**

Among the 17 studies in adults, 11 concluded that PCV20 alone would be cost-saving compared with PCV15, PCV13, PPSV23, PCV15+PPSV23 or no vaccination (including studies with at least one comparison).<sup>23,25,28–30,32–35,38,42</sup> Four studies concluded that PCV20 alone was cost-effective (including studies with at least one comparison).<sup>23,34,37,41</sup> Two studies concluded that PCV20 was not cost-effective (including studies with at least one comparison).<sup>44,46</sup> In two studies in which PCV21 was set as the new vaccine and compared with PCV20, the conclusion was that the new PCV21 vaccine was cost-saving<sup>34</sup> and economically

reasonable.<sup>13</sup> However, in these studies, the results were not simply compared (Table 2).

### **Review of CEA studies of pneumococcal vaccines by country**

The following countries published two or more studies (children or adult): Germany, Japan, South Korea, the UK, and the US. Among the studies from Germany, one in adults showed that PCV20 was cost-saving,<sup>28</sup> and in two pediatric studies of PCV20, one showed that it was not cost-effective<sup>26</sup> while the other concluded that it was cost-saving.<sup>27</sup> In the two pediatric studies, the comparator to PCV20 was different (PCV20 vs. PCV15,<sup>26</sup> and PCV20 vs. PCV13 or PCV15).<sup>27</sup> In all three studies from Japan (one in children<sup>31</sup> and two in adults),<sup>32,33</sup> it was concluded that PCV20 was cost-saving. However, the comparator to PCV20 in each study was different (PCV20 vs. PCV13 or PCV15,<sup>31</sup> PCV20 vs. PPSV23 or PCV15,<sup>32</sup> PCV20 vs. PPSV23).<sup>33</sup> Two studies from South Korea concluded that PCV20 in children was cost-saving<sup>36</sup> and PCV20 in adults was cost-effective.<sup>37</sup> Of the three studies from the UK, one pediatric study concluded that PCV20 was cost-saving compared with other vaccines<sup>40</sup>; however, the two studies in adults came to different conclusions: in one study it was cost-saving (PCV20 vs. PPSV23),<sup>42</sup> while in the other study it was cost-effective (PCV20 vs. no vaccination).<sup>41</sup> In the studies from the US, one pediatric study concluded that PCV20 was cost-saving<sup>43</sup> while the four adult studies arrived at different conclusions: two showed it was not cost-effective (PCV20 vs. no vaccination<sup>44</sup> and PCV20 vs. current recommendation),<sup>46</sup> one showed that PCV20 at ages 50 and 65 years was “potentially economically viable” (PCV20 at ages 50 and 65 years vs. no vaccination or current vaccination),<sup>45</sup> and one showed that PCV21 was more economically favorable than PCV20 (PCV21 vs. PCV20).<sup>13</sup>

### **Overview of analytical framework of CEA studies**

The initial stage of conducting a CEA is to frame the study, i.e., the decision-maker that will use the results, the decision that the decision-maker is facing, and the model estimation.<sup>14</sup> Our guidance covers the analytical framework for CEA as well as how to interpret it based on the reviewed articles. The analytical framework in the reviewed articles included the following main parameters in the reviewed articles: model assumption, comparison, target population, perspective and costing, time horizon, discounting/discount rate, and the parameters of vaccine coverage, vaccine effectiveness, indirect effect, and costs. We also reviewed threshold and uncertainty, biases, as well as the source of funding (Tables 4,5,6 and 7).

### **Model assumption**

If a new vaccine program is introduced, various outcomes are predicted.<sup>11</sup> Therefore, in the analysis, a decision regarding how to model the features of a disease and the impact of interventions is required, which is made by balancing both clinical validity and the purpose of the research.<sup>18,59</sup> All models in a CEA study simplify the real pathways of a disease, according to the research purpose.<sup>18</sup> There are two main approaches to CEA modeling of infectious



**Table 3.** Pneumococcal vaccination schedules in 15 countries.

| Country     | Children (no. of dose)          | Adults                                                                                                                        | Ref.   |
|-------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--------|
| Argentina   | PCV13 (2 + 1)                   | PCV20 ( $\geq 65$ yrs) and specific group (5–64 yrs)                                                                          | 48     |
| Belgium     | PCV13 or PCV15 (3)              | PCV20 ( $\geq 65$ yrs & specific group $\geq 16$ yrs), but PCV20 ( $\geq 65$ yrs) is not funded by the National Health system | 49     |
| Canada      | PCV15 or PCV20 (2 + 1 or 3 + 1) | PCV20 ( $\geq 65$ yrs)                                                                                                        | 50     |
| Denmark     | PCV13 (3)                       | No schedule                                                                                                                   | 49     |
| Germany     | PCV13 or PCV 15 (3)             | PCV20 ( $\geq 60$ yrs), PCV20 (18–59 yrs specific group)                                                                      | 49     |
| Greece      | PCV13 or PCV15 (2 + 1)          | PCV20 ( $\geq 65$ yrs), PCV20 (19–64 yrs specific group)                                                                      | 49     |
| Italy       | PCV*(3)                         | PCV13+PPSV23 ( $\geq 65$ yrs)                                                                                                 | 49     |
| Japan       | PCV20 or PCV15 (3 + 1)          | PPSV23 (65 yrs and 60–64 yrs specific group)                                                                                  | 51, 52 |
| Netherlands | PCV10 (3)                       | PPSV23 ( $\geq 60$ yrs)                                                                                                       | 49     |
| Norway      | PCV13 (3)                       | PPSV23 ( $\geq 65$ yrs) but not funded by National Health System                                                              | 49     |
| South Korea | PCV* (4)                        | PPSV23 ( $\geq 65$ yrs and 19–64 yrs specific group)                                                                          | 53, 54 |
| Spain       | PCV*                            | PPSV23 ( $\geq 65$ yrs and 12–64 yrs specific group)                                                                          | 49     |
| Sweden      | PCV*                            | PPSV23 ( $\geq 75$ yrs and 2–74 yrs specific group)                                                                           | 49     |
| UK          | PCV13 (1 + 1)                   | PPV23 for 65 yrs, PCV13, PPV23 for specific group                                                                             | 55, 56 |
| US          | PCV15 or PCV20 (4)              | PCV15, PCV20, or PCV21 ( $\geq 65$ yrs and 19–64 yrs specific group)                                                          | 57, 58 |

\*Not available detail information.

The definitions of specific groups differ from each country.

PCV: pneumococcal conjugate vaccine, PPSV: pneumococcal polysaccharide vaccine, PPV: pneumococcal polysaccharide vaccine.

diseases: the static model and the dynamic model.<sup>11</sup> In the static model, the force of infection is constant or changes according to individual characteristics, such as age or other individual-dependent factors among vaccinated individuals.<sup>11,14</sup> The static model basically estimates the direct effect of vaccination on vaccinated individuals.<sup>11</sup> In the dynamic model, the force of infection can vary throughout the course of time and depends on disease transmission in the population; direct and indirect effects are also taken into consideration in vaccinated and unvaccinated individuals.<sup>11,14,60</sup> Although it is assumed that the disease course occur randomly in a stochastic (probabilistic) model, this is not assumed in a deterministic model. Both stochastic and deterministic models can be adopted in dynamic and static models.<sup>11,60</sup> A deterministic static model can include decision-tree, Markov-model, and hybrid approaches.<sup>11</sup> The decision-tree approach is a straightforward model for comprehending the options in health-care resourcing and is widely used in economic evaluation.<sup>11,59</sup> The Markov model is another representative form of modeling. Whereas decision-tree models can be used if the timeframe is short and recurring events are uncomplicated, a Markov model may be appropriate for modeling diseases with a particular state in a certain period or that are recurrent.<sup>10,12</sup>

In the reviewed articles, the dynamic model was adopted for only one study in children<sup>40</sup>; the static model was adopted in all other studies.

### Comparison

The concept of “comparison” is the basis of the study question in a CEA study. In the case of pneumococcal disease, pneumococcal vaccines with a higher valency are continually being developed. Let us assume a choice between two alternatives: a new vaccine A and an existing vaccine B. Vaccine A may be in development and existing vaccine B may be a lower-valency vaccine; thus, current standard practice(s) is often used as vaccine B.<sup>14</sup> Because there may be more than two possible vaccines, all relevant alternatives should be considered<sup>10</sup> depending on a particular country’s vaccine strategy or the researcher’s academic interests. A clear description regarding comparators is crucial.

In the reviewed articles in both children and adults, the comparison(s) along with the research question were mostly described clearly in the main text. In some studies, a single comparison was conducted. For example, Kang et al. stated that they aimed “to evaluate the cost-effectiveness of PCV20 compared with PPSV23 in adults aged 65 years and older”<sup>37</sup> and “to evaluate the cost-effectiveness of PCV20 versus PCV13” for the pediatric population in another study.<sup>36</sup> Several studies carried out comparisons; for example, a new vaccine was compared with existing vaccine schedules. It is important to capture the comparisons set in the CEA study according to the research context, as these determine the research outcomes.

**Table 4.** Characteristics of the included studies (children).

| Author (publication year) | Country     | Target population                  | Modeling approach | Time horizon/period       | Discount rate, % | Perspective     | Ref |
|---------------------------|-------------|------------------------------------|-------------------|---------------------------|------------------|-----------------|-----|
| Huang et al. (2024)       | Germany     | Children aged <2 yrs               | Markov            | Lifetime                  | 3                | Societal        | 26  |
| Fridh et al. (2024)       | Sweden      | Children aged <2 yrs               | Markov            | 10-year                   | 3                | Payer           | 39  |
| Rey-Ares et al. (2024)    | Argentina   | Children aged <2 yrs               | Markov            | 10-year                   | 3                | Payer           | 22  |
| Kang et al. (2024)        | South Korea | Children aged 6 weeks to 15 months | Markov            | 10-year/lifetime sequelae | 4.5              | Payer           | 36  |
| Ta et al. (2024)          | Germany     | Children aged <2 yrs               | Markov            | 10-year/lifetime sequelae | 3                | Societal        | 27  |
| Shinjoh et al. (2024)     | Japan       | Children aged <2 yrs               | Markov            | 10-year/lifetime sequelae | 2                | Societal, payer | 31  |
| Rozenbaum et al. (2024)   | US          | Children aged <2 yrs               | Markov            | 10-year/lifetime sequelae | 3                | Societal, payer | 43  |
| Lytle et al. (2023)       | Canada      | Children aged <2 yrs               | Markov            | 10-year/lifetime sequelae | 1.5              | Societal, payer | 24  |
| Wilson et al. (2023)      | UK          | Pediatric population               | Dynamic           | 5-year                    | 3.5              | Payer           | 40  |

Table 5. Characteristics of the included studies (children).

| Author (publication year) | Comparison(s)                                                                                                                            | Vaccine cost (vaccine price, administration cost)                               | Vaccine coverage; % (primary doses/ booster)                | Indirect effect <sup>1</sup> | ICER <sup>2</sup>                                                                                    | Threshold                                 | Uncertainty                 | Funding source | Ref |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------|----------------|-----|
| Huang et al. (2024)       | PCV20 (3 + 1) vs. PCV15 (2 + 1)                                                                                                          | PCV15/20, EUR59.80/66.95, EUR8.00                                               | PCV20: 86/68<br>PCV15: 89/76                                | yes                          | EUR503,620/QALY                                                                                      | EUR50,000, 100,000, 150,000, 200,000/QALY | DSA, PSA, Scenario analyses | MSD            | 26  |
| Fridh et al. (2024)       | PCV20 (2 + 1) vs. PCV15 (2 + 1)                                                                                                          | PCV15/20, SEK653.00/648.69, SEK372.00                                           | –                                                           | yes                          | Dominant                                                                                             | SEK300,000/QALY                           | DSA, PSA, Scenario analyses | Pfizer         | 39  |
| Rev-Ares et al. (2024)    | PCV20 (2 + 1) vs. PCV13 (2 + 1) vs. PCV15 (2 + 1)                                                                                        | PCV13/15/20, USD22.79/22.79/24.53, USD6.14                                      | 78.92                                                       | yes                          | Dominant (both)                                                                                      | USD13,686 (GDP per capita)                | DSA, PSA, Scenario analyses | Pfizer         | 22  |
| Kang et al. (2024)        | PCV20 (3 + 1) vs. PCV13 (3 + 1)                                                                                                          | PCV13/20, USD49.5/54.5, USD15.2                                                 | 97.0                                                        | yes                          | Dominant                                                                                             | USD15,477/KRW20,000,000                   | DSA, PSA, Scenario analyses | Pfizer         | 36  |
| Ta et al. (2024)          | PCV20 (3 + 1) vs. PCV13 (2 + 1) vs. PCV15 (2 + 1)                                                                                        | PCV13/15/20, EUR59.64/59.80/65.44, EUR7.73                                      | 89.9/76.8                                                   | yes                          | Dominant (both)                                                                                      | –                                         | DSA, PSA, Scenario analyses | Pfizer         | 27  |
| Shinjoh et al. (2024)     | PCV20 (3 + 1) vs. PCV13 (3 + 1) vs. PCV15 (3 + 1)                                                                                        | PCV13/15/20, JPY7200/7200/7200, 7200/7200/8102 (2 prices were adopted), JPY4400 | 97.9/97.1                                                   | yes                          | Dominant (both)                                                                                      | USD38,023.39/JPY5 million/QALY            | DSA, PSA, Scenario analyses | Pfizer         | 31  |
| Rozenbaum et al. (2024)   | PCV20 (3 + 1) vs. PCV13 (3 + 1) vs. PCV15 (3 + 1)                                                                                        | PCV13/15/20, USD182.07/181.11/204.59, \$17.63                                   | 89.7/83.5                                                   | yes                          | Dominant (both)                                                                                      | –                                         | DSA, PSA, Scenario analyses | Pfizer         | 43  |
| Lytle et al. (2023)       | PCV20 (2 + 1) vs. PCV13 (2 + 1) vs. PCV15 (2 + 1)                                                                                        | PCV13/15/20, CAD99.91/99.90/109.91, CAD11.00                                    | 84                                                          | yes                          | Dominant (both)                                                                                      | –                                         | DSA, PSA, Scenario analyses | Pfizer         | 24  |
| Wilson et al. (2023)      | 1) PCV20 (1 + 1)<br>2) PCV20 (1 + 1)<br>3) PCV20 (1 + 1)<br>4) PCV20 (2 + 1)<br>5) PCV20 (2 + 1)<br>6) PCV20 (2 + 1)<br>7) PCV20 (2 + 1) | PCV13/15/20, GBP49.10/50.30/56.50, GBP10.06                                     | 95.0/85.0 (year 2006–2010)<br>97.0/91.0 (year 2010 onwards) | no                           | 1) Dominant<br>2) Dominant<br>3) Dominant<br>4) GBP8110<br>5) Dominant<br>6) Dominant<br>7) Dominant | GBP20,000/QALY                            | DSA, Scenario analyses      | Pfizer         | 40  |

<sup>1</sup>‘+’ means ‘followed by.’<sup>1</sup>In base case.<sup>2</sup>ICER: incremental cost-effectiveness ratio.

**Table 6.** Characteristics of the included studies (adults).

| Author (publication year)  | Country     | Target population of vaccination                                                                                                                                                | Modeling approach | Time horizon | Discount rate, %                               | Perspective     | Ref |
|----------------------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------|------------------------------------------------|-----------------|-----|
| Wateska et al. (2024)      | US          | Aged 65 yrs                                                                                                                                                                     | Markov            | Lifetime     | 3                                              | Payer           | 44  |
| Danelian et al. (2024)     | UK          | Aged 65 or 75 yrs                                                                                                                                                               | Markov            | 30-year      | 3.5                                            | Payer           | 41  |
| Nakamura et al. (2024)     | Japan       | Aged 65 years, 60–64 yrs (high risk)                                                                                                                                            | Markov            | Lifetime     | 2                                              | Societal, payer | 32  |
| Altawalbeh et al. (2024)   | US          | At aged 50 and 65 yrs                                                                                                                                                           | Markov            | Lifetime     | 3                                              | Societal        | 45  |
| de Boer et al. (2024)      | Netherlands | Aged 65 yrs                                                                                                                                                                     | Markov            | 15-year      | 4 for costs, 1.5 for benefits                  | Societal        | 34  |
| Kang et al. (2024)         | South Korea | Aged ≥65 yrs                                                                                                                                                                    | Markov            | Lifetime     | 4.5                                            | Payer           | 37  |
| Kühne et al. (2023)        | Germany     | Aged ≥60 yrs, 18–59 yrs (at-risk)                                                                                                                                               | Markov            | Lifetime     | 3                                              | Payer           | 28  |
| Marbaix et al. (2023)      | Belgium     | Aged 18–99 yrs (high risk), 50–99 yrs (moderate risk), 65–84 yrs (low risk)                                                                                                     | Markov            | Lifetime     | 3.0 for costs, 1.5 for benefits                | Payer, patient  | 23  |
| Gourzoulidis et al. (2023) | Greece      | Aged ≥65 yrs, 18–64 yrs (underlying conditions)                                                                                                                                 | Markov            | Lifetime     | 3.5                                            | Payer           | 29  |
| Malene et al. (2023)       | Norway      | Aged ≥65 yrs, ≥18 yrs (risk of pneumococcal diseases)                                                                                                                           | Markov            | Lifetime     | 4 (0–39 yrs), 3 (40–74 yrs), 2 (≥75 yrs)       | Payer           | 35  |
| Restivo et al. (2023)      | Italy       | Aged ≥65 yrs (low risk), 50–100 yrs (at-risk)                                                                                                                                   | Markov            | Lifetime     | 3                                              | Societal        | 30  |
| Cantarero et al. (2023)    | Spain       | Aged ≥60 yrs                                                                                                                                                                    | Markov            | 10-year      | 3                                              | Payer           | 38  |
| Wateska et al. (2023)      | US          | Aged 65 yrs                                                                                                                                                                     | Markov            | Lifetime     | 3                                              | Payer           | 13  |
| Olsen et al. (2022)        | Denmark     | 1) Aged ≥65 yrs and ≥18 yrs (moderate or high risk), ≥50 yrs, ≥65 yrs, ≥65 yrs (moderate or high risk)<br>2) Aged ≥65 yrs and ≥18 yrs (moderate or high risk), ≥50 yrs, ≥65 yrs | Markov            | Lifetime     | 3.5 (0–35 yrs), 2.5 (36–70 yrs), 1.5 (>70 yrs) | Societal        | 25  |
| Hoshi et al. (2022)        | Japan       | Aged 65 yrs                                                                                                                                                                     | Markov            | Lifetime     | 2                                              | Payer           | 33  |
| Mendes et al. (2022)       | UK          | Aged 65–99 yrs, 18–64 yrs (underlying condition)                                                                                                                                | Markov            | Lifetime     | 3.5                                            | Payer           | 42  |
| Smith et al. (2021)        | US          | Aged 65 yrs                                                                                                                                                                     | Markov            | Lifetime     | 3                                              | Payer           | 46  |

### Target population

The target population of a CEA is the group assumed to receive the intervention; thus, in a vaccine program, it is the group assumed to receive the vaccination. The target population can vary according to, e.g., age, sex, occupation, or clinical condition and may also depend on each country's context and history.<sup>14</sup> For CEAs in preventive medicine interventions, such as vaccination programs, a birth cohort or a population-based cohort could potentially be a general population rather than specifically a patient population.

In the pediatric studies reviewed in our literature search, the target populations of vaccine strategies were described as “children under the age of 2 years are directly vaccinated,”<sup>39</sup> or “a new birth cohort entered the model and was vaccinated,”<sup>22</sup> or “a single birth cohort in Germany entered the model.”<sup>26</sup> In the adult studies, the target populations varied by age; for example, Danelian et al. featured a cohort of adults aged either 65 or 75 years old.<sup>41</sup> Other studies had a cohort of 65-year-old adults<sup>13,33,34,44,46</sup> or a population aged 60 or 65 years and older,<sup>37,38</sup> or adults aged ≥18 years.<sup>23,25,28,29,35,42</sup> In those studies with target populations aged ≥18 years, risk profiles by age were taken into account. The target population or target age may be set by a reflection of an existing NIP or to simplify the model scenario of the vaccination schedule in a country's context.

### Perspective and costing

Various decision-makers with specific interests can be assumed in each economic evaluation.<sup>10</sup> The perspective in CEA is directly

related to the costs and consequences counted in the model, e.g., whose costs and consequence are estimated, and to what extent they are evaluated.<sup>10,15</sup> Although there are many potential decision-makers in public health programs, such as a vaccination program, health-care provider, specific payers, and societal perspectives are usually considered.<sup>11</sup> The choice of perspective may depend on the research purpose or research context.<sup>11,14</sup> The perspective of the research relates to the extent of costing and benefits.<sup>14</sup> From the perspective of a health-care provider or payer perspective, the costs directly borne by providers (such as hospitals) or by payers (such as Ministry of Health) are estimated by each.<sup>11</sup> From a societal perspective, the costs incurred by households, such as transportation costs, are included, as well as costs incurred by provider and payer.<sup>11</sup> However, those included costs are not the same from each perspective or in each study. Costs for the vaccination program are estimated by referring to guidelines indicating how to estimate these.<sup>11,14</sup>

Among all studies reviewed, some case analyses were performed from more than one perspective in the base case, e.g., both payer and societal perspectives.<sup>32</sup> Marbaix et al. performed their analysis from three perspectives including the patient perspective as different cost perspectives.<sup>23</sup> In most studies, the perspective is described together with its cost items, to clarify the related costs. Huang et al. described the “societal perspective, including direct medical, direct non-medical, and indirect costs,” with the costs listed in a table.<sup>26</sup> Olsen et al. described a societal perspective but restricted by “... costs of patient time and transportation related to receiving treatment were included, but costs due to productivity loss were excluded,” to clarify the costing.<sup>25</sup> Details of these costs



Table 7. Characteristics of the included studies (adults).

| Author (publication year) | Comparison(s)                                                                                                                                                                                         | Vaccine cost (vaccine price, administration cost)                                                             | Vaccine coverage, %                                                                       | Indirect effect* | ICER                                                                                                                                                                                                                                                                                                                                                                                                 | Threshold                                  | Uncertainty                 | Funding source                                                        | Ref |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-----------------------------|-----------------------------------------------------------------------|-----|
| Wateska et al. (2024)     | PCV20 vs. no vaccination                                                                                                                                                                              | PCV20, USD249.00, USD23.60                                                                                    | Black population 59.8, Non-Black population 70.3,                                         | no               | Black population: USD216,805/QALY<br>Non-black population: USD254,776/QALY                                                                                                                                                                                                                                                                                                                           | USD100,000, 150,000, 200,000, 300,000/QALY | PSA, DSA, Scenario analyses | The National Institute of Allergy and Infectious Diseases             | 44  |
| Danelian et al. (2024)    | PCV20 vs. no vaccination, (PPSV23 vs. no vaccination)                                                                                                                                                 | PPSV23/PCV20, GBP16.80/56.80, GBP10.06                                                                        | 70                                                                                        | no               | – PCV20 vs. no vaccination: GBP14,865/QALY (65 yrs), GBP13,223/QALY (75 yrs)<br>– PPSV23 vs. no vaccination: GBP13,309/QALY (65 yrs), GBP12,326/QALY (75 yrs)<br>PCV20, likely to avert more case of PD.                                                                                                                                                                                             | GBP20,000, 30000/QALY                      | PSA, Scenario analyses      | UK Health Security Agency and UK Department of Health and Social Care | 41  |
| Nakamura et al. (2024)    | PCV20 vs. PPSV23 vs. PCV15                                                                                                                                                                            | PPSV23/PCV15/PCV20, JPY4735/7200/8102, JPY3250                                                                | 40.4                                                                                      | no               | Dominant (both, all perspective)                                                                                                                                                                                                                                                                                                                                                                     | JPY5,000,000/QALY                          | DSA, PSA, Scenario analyses | Pfizer                                                                | 32  |
| Altawalbeh et al. (2024)  | 5 strategies were evaluated<br>1) PCV20 at ages 50 and 65 yrs<br>2) PCV15+PPSV23 at ages 50 and 65 yrs<br>3) Current PCV20 <sup>1</sup><br>4) Current PCV15 +PPSV23 <sup>1</sup><br>5) No vaccination | PCV/PPSV, USD226.43/117.08, USD23.6<br>≥65 yrs, 59.8<br>Non-Black population 50–64 yrs, 38.8<br>≥65 yrs, 70.3 | Black population 50–64 yrs, 38.8<br>Non-Black population 50–64 yrs, 38.8<br>≥65 yrs, 70.3 | no               | Black population/non-balck population – PCV20 <sup>2</sup> vs. PCV15+PPSV23 (strategy 2), less cost, less effective (both)<br>– PVC20 <sup>2</sup> vs. current PCV20 (strategy 3), dominant/approximately USD42,000/QALY (calculated from table)<br>– PVC20 <sup>2</sup> vs. current PCV15+PPSV23 (strategy 4), dominant (both)<br>– PCV202 vs. no vaccination (strategy 5), dominant/USD62,083/QALY | USD100,000, 150,000/QALY                   | DSA, PSA                    | National Institute of Allergy and Infectious Diseases                 | 45  |
| de Boer et al. (2024)     | 7 strategies were evaluated<br>1) No vaccination<br>2) 3xPPSV23<br>3) PCV15<br>4) PCV20<br>5) PCV15 + 3xPPSV23<br>6) PCV20 + 3xPPSV23<br>7) PCV21                                                     | PPSV23/PCV15/PCV20/PCV21, EUR25.94/74.73/82.17/82.17, EUR21.00                                                | 70                                                                                        | yes              | – PCV20 vs. PCV15, PPSV23(x3), PCV15, PCV15+PPSV23(x3) (with PCV10 in children), dominant<br>– PCV20 vs. no vaccination (with PCV10/PCV13/PCV15/PCV20 in children), EUR9,051/QALY, EUR10,228/QALY, EUR11,173/QALY, EUR22,550/QALY<br>– PCV20+PPSV23(x3) vs. PCV20, EUR81,193–115,412/QALY<br>– PCV20 vs. PCV21, PCV21, dominant                                                                      | EUR20,000/QALY                             | DSA, PSA                    | Netherlands Ministry of Health, Welfare and Sport                     | 34  |

(Continued)

Table 7. (Continued).

| Author<br>(publication year)  | Comparison(s)                                     | Vaccine cost<br>(vaccine price,<br>administration<br>cost)                  | Vaccine<br>coverage,<br>%<br>Low/Mild/<br>High risk                                                                                                                                           | Indirect<br>effect* | ICER                            | Threshold      | Uncertainty                       | Funding source | Ref |
|-------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------|----------------|-----------------------------------|----------------|-----|
| Kang et al. (2024)            | PCV20 vs. PPSV23                                  | PCV20/PPSV23,<br>USD57.5/19.6,<br>USD15.1                                   | 65–74 yrs,<br>59.6/63.1/<br>64.1<br>75–84 yrs,<br>59.4/62.2/<br>62.8<br>85–99 yrs,<br>59.4/62.2/<br>62.8                                                                                      | yes                 | USD2,677/QALY                   | USD16,824/QALY | DSA, PSA,<br>Scenario<br>analyses | Pfizer         | 37  |
| Kühne et al. (2023)           | PCV20 vs. Current<br>strategy <sup>3</sup>        | PPSV23/PCV13/<br>PCV15/PCV20,<br>EUR33.88/72.50/<br>76.74/76.74,<br>EUR8.19 | Low/Mild/<br>High risk<br>18–49 yrs,<br>0.0/7.1/7.1<br>50–59 yrs,<br>0.0/13.0/<br>13.0<br>60–64 yrs,<br>13.3/32.0/<br>32.0<br>65–74 yrs,<br>41.0/35.2/<br>35.2<br>≥75 yrs, 41.0/<br>35.2/35.2 | yes                 | Dominant                        | –              | DSA, PSA,<br>Scenario<br>analyses | Pfizer         | 28  |
| Marbaix et al. (2023)         | PCV20 vs. 1) no<br>vaccination<br>2) PCV15+PPSV23 | PCV20/PCV15<br>+PPSV23,<br>EUR66.91/108.21                                  | Low/Mild/<br>High risk<br>18–49 yrs,<br>0.19/0.14/<br>6.00<br>50–64 yrs,<br>0.81/6.00/<br>7.20<br>65–74 yrs,<br>8.04/15.19/<br>10.96<br>75–84 yrs,<br>9.00/17.94/<br>11.97                    | yes                 | 1) EUR4,164/QALY<br>2) Dominant | EUR35,000/QALY | DSA, PSA,<br>Scenario<br>analyses | Pfizer         | 23  |
| Gourzoulidis et al.<br>(2023) | PCV20 vs. PCV15<br>vs. PCV15+PPSV23               | PCV20/PPSV23/<br>PCV15, EUR 70.97/<br>31.45/72.32                           | –                                                                                                                                                                                             | no                  | Dominant (both)                 | EUR34,000/QALY | DSA, PSA,<br>Scenario<br>analyses | Pfizer         | 29  |
| Malene et al. (2023)          | PCV20 vs. PPSV23                                  | PCV20/PPSV23,<br>EUR77.08/26.62,<br>EUR46.96                                | 75                                                                                                                                                                                            | yes                 | Dominant                        | –              | DSA, PSA,<br>Scenario<br>analyses | Pfizer         | 35  |

(Continued)

Table 7. (Continued).

| Author<br>(publication year) | Comparison(s)                                        | Vaccine cost<br>(vaccine price,<br>administration<br>cost)                   | Vaccine<br>coverage,<br>%<br>Low risk/at-<br>risk,<br>65/25                                                                                                                                                   | Indirect<br>effect* | Dominant                                                                                                                                                | ICER | Threshold                           | Uncertainty                       | Funding source                                                                                                                                                                 | Ref |
|------------------------------|------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Restivo et al. (2023)        | PCV20 vs. PCV15<br>+PPSV23                           | PCV/PPSV23,<br>EUR44.63/19.13,<br>EUR6.80 (per<br>injection)                 | Low risk/at-<br>risk,<br>65/25                                                                                                                                                                                | no                  | Dominant                                                                                                                                                |      | EUR40,000/QALY                      | DSA, PSA,<br>Scenario<br>analyses | MSD                                                                                                                                                                            | 30  |
| Cantarero et al.<br>(2023)   | PCV20 vs. PCV15<br>+PPSV23                           | PCV15/PCV20/<br>PPSV23,<br>EUR55.50/49.76/<br>14.29, EUR6 (per<br>injection) | Low/Mod/<br>High risk <sup>4</sup><br>18–49 yrs,<br>0.0/0.0/0.0<br>50–64 yrs,<br>1.3/15.1/<br>59.1<br>65–74 yrs,<br>6.6/31.1/<br>48.3<br>75–84 yrs,<br>35.6/46.1/<br>66.9<br>85–99 yrs,<br>69.5/60.7/<br>77.0 | no                  | Dominant                                                                                                                                                |      | EUR25,000/QALY                      | DSA, PSA,<br>Scenario<br>analyses | Pfizer                                                                                                                                                                         | 38  |
| Wateska et al.<br>(2023)     | PCV21 vs. PCV20                                      | PCV20/PCV21,<br>USD249.00/<br>333.00, USD23.60                               | Black<br>population<br>59.8,<br>Non-black<br>population<br>70.3                                                                                                                                               | both                | (black population/non-black<br>population) approximately<br>USD8,000/QALY / USD40,000/QALY<br>(calculated from table)<br>PCV21, economically reasonable |      | USD100,000–150,000,<br>200,000/QALY | DSA, PSA,<br>Scenario<br>analyses | –                                                                                                                                                                              | 13  |
| Olsen et al. (2022)          | 1) PPSV23+PCV20<br>vs. PPSV23<br>2) PCV20 vs. PPSV23 | PPSV23/PCV20,<br>EUR39.7/120.7,<br>EUR40.0 (per<br>injection)                | 73, 25<br>(booster)                                                                                                                                                                                           | yes                 | Dominant (both)                                                                                                                                         |      | –                                   | DSA, PSA                          | Pfizer                                                                                                                                                                         | 25  |
| Hoshi et al. (2022)          | PCV20 vs. PPSV23                                     | PPSV23/PCV20:<br>USD42.9/172.996,<br>USD27.7                                 | 40.4                                                                                                                                                                                                          | no                  | Dominant                                                                                                                                                |      | –                                   | DSA                               | Emerging and Re-emerging Infectious<br>Diseases, Health and Labour Sciences<br>Research Grants from the Ministry of Health,<br>Labour and Welfare of Japan and JSPS<br>KAKENHI | 33  |

(Continued)

Table 7. (Continued).

| Author<br>(publication year) | Comparison(s)                                                                                                                                                | Vaccine cost<br>(vaccine price,<br>administration<br>cost) | Vaccine<br>coverage,<br>%<br>Mod/High<br>risk:<br>18–64 yrs,<br>43.4/37.5<br>Low/Mod/<br>High:<br>65–74 yrs,<br>42.9/79.0/<br>83.7<br>75–84 yrs,<br>73.3/89.6/<br>95.5<br>85–99 yrs,<br>77.9/85.9/<br>85.7 | Indirect<br>effect* | Dominant                                                                                                                                                                    | ICER | Threshold<br>GBP20,000–30,000/<br>QALY | Uncertainty<br>PSA,<br>Subgroup<br>analysis | Funding source                                           | Ref |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------------------------------------|---------------------------------------------|----------------------------------------------------------|-----|
| Mendes et al. (2022)         | PCV20 vs. PPSV23                                                                                                                                             | PPSV23/PCV20,<br>GBP16.80/56.50,<br>GBP10.06               |                                                                                                                                                                                                            | yes                 | Dominant                                                                                                                                                                    |      |                                        |                                             | Pfizer                                                   | 42  |
| Smith et al. (2021)          | 6 strategies were<br>evaluated<br>(1) Current<br>recommendation <sup>5</sup><br>(2) PPSV23<br>(3) PCV15<br>(4) PCV15+PPSV23<br>(5) PCV20<br>(6) PCV20+PPSV23 | PPSV23/PCV20,<br>USD105/222,<br>USD25.84                   | 63.6                                                                                                                                                                                                       | both                | - PCV20 vs. Current recommendation,<br>approximately USD172,000/QALY<br>- PCV20+PPSV23 vs. PCV20,<br>>USD3.1 million/QALY<br>(no childhood vaccination indirect<br>effects) |      | USD100,000–150,000/<br>QALY            | PSA, DSA,<br>Scenario<br>analysis           | National Institute of Allergy and Infectious<br>Diseases | 46  |

\*+ means 'followed by'.

ICER: incremental cost-effectiveness ratio.

\*In base case.

<sup>1</sup>For all ≥65 yrs and for younger adults with a high risk of pneumococcal disease.<sup>2</sup>At both ages 50 and 65 years (strategy 1).<sup>3</sup>Age/risk-dependent use of PPSV23 or PCV13+PPSV23 with PPSV23 revaccination.<sup>4</sup>The vaccine efficacy was shown in all age groups.<sup>5</sup>PPSV23+optional PCV13 after clinical shared decision making.

are included in the main text or other materials in the published studies. The perspective and cost details of a CEA should be reported and carefully captured.

### Time horizon

The CEA analysis is conducted at a single point ahead of a new intervention. However, the pathway of disease in a model is usually considered over a certain period of time. The time horizon (i.e., the period over which the costs and consequences of program intervention are measured and evaluated in an economic evaluation<sup>59</sup>) can be set according to the length of the disease course. An appropriate time horizon should be sufficiently long to reveal the difference in some cases between the new and existing vaccine in terms of the costs and consequences evaluated.<sup>11</sup> Depending on the model assumption adopted or the features of the disease, the life course may be an appropriate time horizon.<sup>10</sup>

In the pediatric studies reviewed herein, the most common time horizon was 10 years; only one study had the span of a lifetime.<sup>26</sup> It is notable that, in some studies, although the time horizon was set over several years, long-term sequelae were considered over a lifetime.<sup>24,27,31,36,43</sup> In the articles reviewed in this study, the time horizons differed for adults and children. In adult studies, although the most common time horizon was up to age 100 years or a lifetime, spans of 10, 15, and 30 years were used in some studies.<sup>34,38,41</sup>

### Discounting/discount rate

In CEA modeling, the future costs and consequences of an intervention are calculated at their present value. However, it is not possible to sum up the costs and benefits that may occur at different time points. For example, a sum of money deposited or invested now may show a return in the distant future, but an earlier benefit is preferred. This is the so-called “time preference” concept in economics.<sup>10</sup> In a health-care setting, although program costs occur in the present, the disease pathway can vary. Therefore, the parameter of discounting is considered in the model, depending on the specific features of the disease or intervention. Whereas vaccination programs, such as those for influenza may, have immediate short-term effects, the effectiveness of a vaccination program for pneumococcal disease may last for a relatively long time.<sup>14</sup> In the latter case, future costs and consequences are reduced by discounting as the benefits of the vaccination program become apparent.<sup>10</sup> WHO guide recommends the rate in the guideline in the country.<sup>14</sup> The following two analysis rates are also recommended: 3% and 0% discounting for costs and health outcomes, respectively, and 3% discounting for both costs and health outcomes.<sup>14</sup> Alternatively, in the absence of national guidelines, a discount rate of 3–5% would be common.<sup>10,11</sup>

All the reviewed studies identified in our literature search indicated the discounting rate, which ranged from 1.5% to 4.5% per year according to cited references or guidelines. In all pediatric studies, the same discount rate was adopted for both costs and health outcomes. This was

also the case in most adult studies; however, de Boer et al. in the Netherlands and Marbaix et al. in Belgium adopted different rates for costs and health outcomes,<sup>23,34</sup> and Malene et al. and Olsen et al. adopted different rates for different age groups.<sup>25,35</sup>

### Parameters

Parameters input in the model should be estimated based on the best existing data that are available from literature reviews on studies such as clinical trials and observational studies, or from appropriate sources.<sup>11</sup>

### Vaccine coverage

One feature of CEAs for public health programs is the inclusion of healthy populations in the model. Vaccine coverage, including vaccine compliance, may relate to vaccine effectiveness and herd immunity.<sup>14</sup> Officially validated data could be a source of vaccination coverage information.<sup>11,14</sup> In a scenario where some healthy individuals are vaccinated via a public health program and some are not, the vaccine coverage may affect the calculated cost estimate and effectiveness. Therefore, the extent of vaccine coverage assumed in the model should be clearly stated.<sup>14</sup>

Among the studies reviewed herein, most indicated the vaccine coverage. In the pediatric studies, the vaccine coverage was assumed to be as high as 78.92% (completed vaccine schedule) in Argentina,<sup>22</sup> 97.0% in South Korea,<sup>36</sup> and 97.9% (for the primary dose) in Japan.<sup>31</sup> In some articles, the number of doses assumed to be received was different; for example, Ta et al. reported a vaccine coverage of 89.9% and 76.8% for the primary and booster dose, respectively.<sup>27</sup>

In some adult studies, the reported values input differed depending on the targeted age range or study setting. In some adult studies, the assumed vaccine coverage was over a wide range, and different values were input depending on the disease risk.<sup>23,28,30,37,38,42</sup>

### Vaccine effectiveness

Vaccine effectiveness is an important parameter affecting outcomes; this can be estimated from well-designed randomized controlled trials in a CEA.<sup>10–12</sup> In CEAs where an unlicensed vaccine is being evaluated, the evidence for vaccine effectiveness may be limited. Vaccine effectiveness data are estimated from clinical or epidemiology study data; however, these data might not be directly matched to the research purpose of a CEA.<sup>10</sup>

In some articles, we reviewed, the vaccine effectiveness of PCV20 was assumed to be that of another PCV, such as PCV13. In some pediatric studies,<sup>22,24,27,39</sup> the effectiveness of PCV20 against invasive pneumococcal disease was based on Savulescu et al.’s study.<sup>61</sup> In adult studies,<sup>25,28,32,35,37,38,41,42,45</sup> the vaccine effectiveness of PCV20 against invasive pneumococcal disease and community-acquired pneumonia referred to the study by Bonten et al.<sup>62</sup> QALYs, which are the health-related outcomes in a CEA, can be also derived with appropriate information on health-related quality of life and life-years.<sup>10,11</sup>



### Indirect effects

The ultimate aim of a vaccination program is to prevent vaccinated individuals from becoming infected with an infectious disease. Moreover, via vaccination, individuals can avoid acquiring infectious diseases from their surrounding environment. The spread of diseases among non-vaccinated individuals may also be reduced owing to the achievement of herd immunity.<sup>11,14</sup> As a feature of the economic evaluation of vaccination, these effects can be considered indirect effects in a CEA model. In the static model, risk reduction from herd immunity may be modeled as an effect.<sup>11</sup> In addition to herd immunity, the disease burden owing to factors such as serotype replacement or antimicrobial resistance can also be considered indirect effects in the model; these effects are described as indirect effects.<sup>14</sup>

Of the nine pediatric studies reviewed herein, eight considered indirect effects to mean the effects among unvaccinated individuals in the base case model.<sup>22,24,26,27,31,36,39,43</sup> Of the 17 adult studies we reviewed, nine modeled indirect effects in the unvaccinated group in the base case.<sup>13,23,25,28,34,35,37,42,46</sup> In some studies, herd immunity was described as an indirect effect. However, in a pediatric study, Huang et al. used serotype replacement in addition to herd protection as indirect effects in a disease pathway.<sup>26</sup> In an adult study, de Boer et al. reported “taking into account indirect protection and serotype replacement from a switch to higher-valency PCVs in childhood vaccination” were indirect effects owing to serotype replacement over a certain period and the decreased incidence among older adults owing to childhood vaccine serotypes added to PCV10.<sup>34</sup> These effects in a CEA model can influence the effectiveness of the model and the outcome of the CEA; therefore, it is crucial to accurately capture such information.

### Costs

Costs are based on the concept of “opportunity cost,” which refers to the value of what is given up by others.<sup>10</sup> Costs for vaccination and costs to be averted that are related to each treatment are estimated based on the disease pathway or alternatives in the model. Because the price of a vaccine that is still in development or not yet licensed is unavailable, this cost would be based on an assumption or with reference to the price of an existing vaccine.<sup>11</sup> Medical treatment refers to costs related to the fact that vaccines may reduce the probability of disease in the model; this information may be obtained from the literature. Therefore, it is important to state which costs are input in the model and to cite their source, as well as the rationale for using the source.<sup>11,14</sup> The vaccine price includes uncertainty if it is not known. As such, the uncertainty of costs within some ranges can be explored in several ways, such as in threshold analysis and one-way sensitivity analysis.<sup>11,14</sup>

In some pediatric studies, we reviewed, the vaccine cost was adopted using vaccine prices from publicly available data, e.g., available price lists.<sup>26,27,36,39,40</sup> In some studies, a different vaccine price (i.e., PCV13), sometimes weighted, was adopted.<sup>22,31,43</sup> In adult studies, both approaches were also used. Hoshi et al. and Nakamura et al. used the adult vaccine price according to the US Centers for Disease Control and

Prevention; the prices for PCV20 and PCV15 were unavailable in Japan at the time of study.<sup>32,33</sup> In some studies, costs associated with side effects or visits to a doctor were counted as costs related to vaccination.<sup>13,23,44–46</sup>

In most reviewed studies, vaccine costs were shown as both the vaccine price and administration fees. In one study, only the cost of the vaccine was shown.<sup>29</sup> The treatment cost of pneumococcal disease was calculated differently in each study setting, i.e., according to disease pathway, age, or severity of disease.

### Threshold and uncertainty

If the results for the ICER are below the threshold, the new vaccine program is deemed to be cost-effective; if the ICER is above the threshold, the new vaccine program is not considered to be cost-effective. In some articles reviewed in our study, the threshold was explicit, with citation of country-specific guidelines. For instance, two studies among adults in the UK adopted the same threshold range: GBP 20,000–30,000/QALY,<sup>41,42</sup> referring to a report of the National Institute for Health and Care Excellence. In some studies in both children and adults, the researchers did not indicate a clear threshold in their study.<sup>24,25,27,28,33,35,43</sup>

Every CEA includes some uncertainty, and accounting for this uncertainty is crucial with regard to which parameter(s) are used in the model or how changing one or more parameters may affect the outcome.<sup>10,11</sup> To assess uncertainty, a DSA and PSA were performed in most pediatric and adult studies we reviewed; moreover, scenario analyses were performed in some studies. As an example in an adult study, Cantarero et al. included work absenteeism owing to illness in a scenario analysis from a societal perspective, where the related costs were not considered in the base case from a payer perspective.<sup>38</sup>

### Biases

A CEA is a model-based study that relies on the parameters input to the model. Therefore, some bias may be present in the model assumptions or parameters estimated.<sup>11</sup> The funding sources in economic evaluation should be reported. Among the studies reviewed herein, many (19/26) were sponsored by pharmaceutical companies, including all nine pediatric studies. Because sponsorship bias can occur in CEA studies,<sup>63</sup> it is important to be fully transparent regarding a study's funding source.<sup>11</sup>

### Conclusion

In this article, we conducted a narrative literature review of recent CEA studies involving the PCV20 vaccine to provide an overview of the CEA models used and to offer guidance on how to interpret CEA studies in vaccination programs. Despite differences in the model structure and parameters with respect to the research setting in

the reviewed studies, the PCV20 vaccine was relatively favorable for both pediatric and adult indications.

New PCVs with higher valency are continually being developed. The results of a CEA may inform decisions regarding whether to introduce a new vaccine or replace existing vaccines in a vaccine schedule. Thus, additional CEA studies on PCVs will likely be performed and published as new, higher valencies are developed. However, as stated previously, a CEA may be variable and will contain a level of uncertainty. It is therefore important to interpret the results of a CEA study with an understanding of the features and limitations of the analysis.

## Author contributions

A.S.: Conceptualization, Research, Writing – Original draft preparation, SL.H.: Research, Reviewing and Editing M.K.: Conceptualization, Reviewing and Editing.

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## Data availability statement

Data openly available in a public repository that issues datasets with DOIs.

## Ethical statement

This research is not human subjects research.

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